

Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!**

Directions: For each problem construct an integral which solves the given problem. Do **NOT** expend time evaluating the constructed integrals

1. (10 pts) Find the area of the bounded region bounded between the curves

$$y = f(x) = x^2 + 4x, y = g(x) = 2x + 3$$

2. (15 pts) Let R be the region in the first quadrant bounded between the curves

$$y = f(x) = \frac{5}{1+x}, x\text{-axis}, 0 \leq x \leq 4$$

- A. Find the volume of the solid of revolution generated by revolving R about the x -axis
- B. Find the volume of the solid of revolution generated by revolving R about the y -axis

3. (10 pts) Find the area enclosed in one loop of the lemniscate given by

$$r^2 = 5 \sin 2\theta$$

4. (10 pts) Find the length of the curve given by $y = f(x) = xe^{-x}$, $0 \leq x \leq 3$.

5. (10 pts) Find the area of the region bounded between the curves

$$y = f(x) = 3x - x^2, y = g(x) = x - 3, -3 \leq x \leq 3$$

6. (15 pts) Let R be the bounded region in the first quadrant bounded between the curves

$$y = f(x) = \frac{x^3}{9}, y = g(x) = x$$

- A. Find the volume of the solid of revolution generated by revolving R about the x -axis
- B. Find the volume of the solid of revolution generated by revolving R about the y -axis

7. (15 pts) Find the surface area of the surface generated by revolving the arc given by

$$y = f(x) = \tan x, 0 \leq x \leq \pi/4 \text{ about}$$

A. x -axis

B. y -axis

8. (10 pts) Find the area of the region bounded between the curves

$$x = y - 2, x = 2y - y^2$$

9. (15 pts) Let R be the bounded region in the first quadrant bounded between the curves

$$x^2 + y^2 = 9, x + y = 3$$

A. Find the volume of the solid of revolution generated by revolving R about the line $y = -2$

B. Find the volume of the solid of revolution generated by revolving R about the line $x = 7$